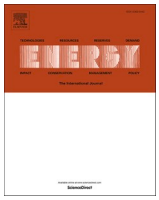
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Optimization and economic evaluation of carbon dioxide BOG Re-liquefaction process considering storage pressure and nitrogen impurity

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ABSTRACT

The increasing demand for carbon capture and storage (CCS) has heightened the need for liquefied carbon dioxide (LCO_2) carriers, which require re-liquefaction systems to manage boil-off gas (BOG) generated during transportation. This study proposed new two-stage separation re-liquefaction processes with mixed refrigerants for CO_2 BOG containing nitrogen impurity. The proposed processes were evaluated and compared with conventional processes based on annual CO_2 emissions under different storage pressure conditions of 6.5 bar and 15 bar. The results demonstrated that the proposed processes reduced annual CO_2 emissions by up to 71.6 % and 35.6 % at the storage pressure of 6.5 bar and 15 bar, respectively. Furthermore, the economic feasibility was assessed by considering annual CO_2 emissions and annual transported CO_2 amount under both pressure conditions. The results indicated that the cases of storage pressure at 6.5 bar resulted in lower costs due to reduced tank costs and the higher density of LCO_2.

1. Introduction

The global carbon dioxide (CO_2) emissions reached 35 billion tons per year in 2020, spurring growing interest in carbon capture and storage (CCS) [1]. CCS has emerged as a key technology for reducing CO_2 emissions and has already been commercially operated on land for decades [2]. In particular, International Renewable Energy Agency expects that CCS could contribute to approximately 20 % of CO_2 emissions reductions by 2050 [3].

The demand for large-scale CO_2 carriers has increased as the capacity of captured and stored CO_2 grows [4]. Options for transporting captured CO_2 to storage sites include pipelines and shipping vessels, with ships reported to be more economical, particularly long-distance transportation [5]. One of the CCS projects, the Northern Lights Project, is operating a 7500 m^3 ship with the goal of injecting 1.5 to 5 million ton of CO_2 per year [6]. HD Hyundai has been constructing two 22,000 m^3 liquefied CO_2 (LCO_2) transport ships [7]. The need for larger CO_2 transport ships continues to grow, with ships ranging from 40,000 to 100,000 m^3 being proposed [8,9].

There have been studies of different LCO_2 storage pressures because it is advantageous to transport CO_2 as a liquid for large-scale transport and CO_2 cannot be liquefied at atmospheric pressure. Pure CO_2 can be liquefied and stored at various pressure conditions between the triple point (5.1 bar, −56.6 °C) and the critical point (73.8 bar, 31.1 °C) [10,11]. In particular, previous studies have demonstrated that storage pressures above 25 bar are economically unfeasible when considering the economics of various pressure conditions in the CCS value chain, which includes CO_2 liquefaction systems, storage tanks, and CO_2 transport ships [11]. The most commonly mentioned storage pressure conditions below 25 bar are 6.5–7.0 bar [10–20] and 15 bar [10–13,17,21]. At a storage pressure of 6.5 bar, the amount of LCO_2 that can be stored per unit volume can be increased by approximately 8.5 % compared to a storage pressure of 15 bar, due to the difference in density [16,17]. Conversely, at a storage pressure of 15 bar, the higher liquefaction temperature reduces the power consumption required for liquefaction compared to 6.5 bar [10,11]. Table 1 shows that most studies have assumed storage pressures of 6–7 bar and 15 bar.

Because the CO_2 liquefaction process consumes significant energy,

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Table 1
Previous studies on CO_2 liquefaction and CO_2 BOG re-liquefaction system.

Literature	Feed CO_2 concentration (mol %)	Storage pressure (bar)	Refrigerant
CO_2 Liquefaction			
Alabdulkarem A et al. [22]	99.93	6	CO_2, NH_3, Propane, R134a
Deng H et al. [10]	96.86–98.42	7–70	NH_3
Seo Y et al. [12]	100	6, 15	NH_3, propane, R134a
Aliyon K et al. [23]	97	19.7	CO_2, NH_3
Seo Y et al. [11]	100	6–65	CO_2, NH_3, propane, ethene
Jackson S et al. [13]	100	7, 8, 15	CO_2, NH_3
Chen F & Morosuk T [21]	100	15	NH_3, Propane, R134a
Lee S et al. [15]	94.4	6.5	CO_2
CO_2 BOG re-liquefaction			
Chu B et al. [16]	98.386	6.2	Propane, Ethylene
Lee Y et al. [20]	64	7	CO_2
Lee S et al. [14]	100	6.5	CO_2
Lee J et al. [17]	100	7, 12, 15, 26	CO_2, NH_3, Propane, Ethane
Yoo B [18]	100	6.59	LNG cold energy
Lu J et al. [19]	100	6.59	NH_3, R22, R23

many studies have been done on the optimal process design for CO_2 liquefaction. The required liquefaction temperatures for pure CO_2 at storage pressures of 6.5–7 bar and 15 bar are −50.6 to −48.5 °C and −27.7 °C, respectively [15,17]. Therefore, previous studies have proposed liquefaction processes utilizing single refrigerants such as ammonia, propane, CO_2, propylene, and R134-a to meet these liquefaction temperatures (Table 1). Among these, the liquefaction process using ammonia as a refrigerant has demonstrated the lowest power consumption and is regarded as the most efficient liquefaction process [12,22,23]. However, most of the previous CO_2 liquefaction studies focus on CO_2 liquefaction, rather than CO_2 boil off gas (BOG) re-liquefaction.

To transport LCO_2 over long distances by ship, the CO_2 BOG re-liquefaction system is essential as part of the cargo handling system (CHS). LCO_2 must be stored at temperatures below room temperature, which can lead to BOG problems when transported by ship [24]. CO_2 BOG can be managed in three ways: venting it while paying the carbon tax, re-liquefying it or holding it at the expense of increased tank pressure. Venting BOG not only releases greenhouse gas (GHG) emissions but also results in cargo loss. Additionally, when considering the carbon tax, venting may be less economical compared to re-liquefaction. When the voyage is short, allowing the tank pressure to rise may be a practical option. Previous study has shown that the holding time of LCO_2 storage tank is approximately 12–13 days when the pressure increases from 6 bar to 8 bar [25]. For longer voyage exceeding this duration, CO_2 BOG re-liquefaction becomes essential to maintain operational safety and prevent cargo loss. Previous studies on CO_2 BOG re-liquefaction, similar to those on CO_2 liquefaction, have proposed re-liquefaction processes using a single refrigerant such as ammonia, propane, or CO_2 (Table 1). For liquefied natural gas (LNG)-powered ships, CO_2 BOG re-liquefaction using the cold heat of LNG BOG has also been suggested [18].

However, most previous CO_2 BOG re-liquefaction studies have assumed pure CO_2, while CO_2 BOG may contain impurities introduced during the capture process [10,15,16,23]. Specifically, impurities with lower boiling points than CO_2, such as nitrogen (N_2), will rapidly vaporize within the LCO_2 storage tank [16,26]. Consequently, the CO_2 BOG will turn into a mixture with a higher concentration of N_2 compared to LCO_2 [16,26]. Previous experiments have shown that higher N_2 content in CO_2 results in lower required temperature for liquefaction compared to pure CO_2 [27]. Therefore, when designing the CO_2 BOG re-liquefaction process, it will be essential to account for the N_2 content in the LCO_2.

Although some studies on CO_2 BOG re-liquefaction processes consider impurities, they propose large and complex systems which are unsuitable for ships that require compact design. Additionally, the re-liquefaction efficiency in these systems is often low. Chu B et al. proposed a cascade re-liquefaction process using propane-ethylene cooling the BOG to −65.6 °C to achieve a re-liquefaction rate of over 98 % [16]. However, this process involved multiple separate compression trains in the cascade cycle and did not account for dry ice formation. Lee Y et al. proposed a re-liquefaction process using a distillation column to remove N_2 from CO_2 BOG, but it was not only more complex than the conventional re-liquefaction process, but also increased energy consumption by more than 1.5 times [20].

Using mixed refrigerants to liquefy a mixture is more efficient than using a single refrigerant and does not complicate the process. The PRICO Single Mixed Refrigerant (SMR) liquefaction process for natural gas (NG) liquefaction serve as a representative example [28–35]. This process has demonstrated higher heat exchange efficiency compared to single refrigerant processes when liquefying NG [34]. Additionally, the SMR process does not require the addition of a compression train like a cascade cycle and is structurally similar to processes that use a single refrigerant. However, there are no studies yet that have optimized and proposed the composition of mixed refrigerants for CO_2 BOG.

In addition, previous CO_2 BOG re-liquefaction studies did not consider their contribution to reducing GHG emissions, focusing instead on evaluating compression energy consumption or re-liquefaction rate [14,16–20]. The primary purpose of recently discussed large CO_2 carriers is to mitigate global warming by reducing CO_2 emissions. Evaluating the process based solely on compression energy consumption may overlook CO_2 purged along with non-condensable gases. Additionally, evaluating the process based solely on the re-liquefaction rate may lead to excessive compression energy consumption, which in turn may result in increased CO_2 emissions to generate electricity. Thus, both the amount of CO_2 purged and generated by the power consumed in the re-liquefaction process must be considered, which can be represented by the overall CO_2 emissions of the re-liquefaction process. Therefore, evaluating the re-liquefaction process based on its annual CO_2 emissions would provide a more accurate view of its environmental impact.

This study proposes a new CO_2 BOG re-liquefaction process considering N_2 impurity by applying mixed refrigerants and two-stage separation method to reduce net CO_2 emissions. Two storage pressures, 6.5 and 15 bar, are considered because they are the most commonly studied pressure conditions in previous studies. Additionally, this study determines the cost-effective storage pressure through an economic evaluation of the proposed process, considering net CO_2 emissions.

2. Backgrounds

2.1. Liquefaction process

Fig. 1(a) shows the vapor compression cycle using a single refrigerant, which was conventionally used in previous CO_2 liquefaction studies [11–13,17,23]. A single refrigerant was compressed in compressor 2 and cooled in cooler 2 by cooling water. After undergoing Joule-Thomson (JT) expansion in valve 2, single refrigerant absorbed heat from the target gas in the heat exchanger and then circulated back to compressor 2. The target gas entered the compressor 1, where it was compressed to high pressure and cooled to a low temperature through heat exchange with the refrigerant in the heat exchanger. It then underwent JT expansion through valve 1 to the pressure of the storage tank, where it was partially liquefied. The liquid separated in the separator flowed into the storage tank and the non-condensable gas was either purged or recycled into the compressor. If the CO_2 was pure, all of the separated gas could be recycled back into the compressor. However, if the CO_2 contained light impurities, purging was necessary to prevent

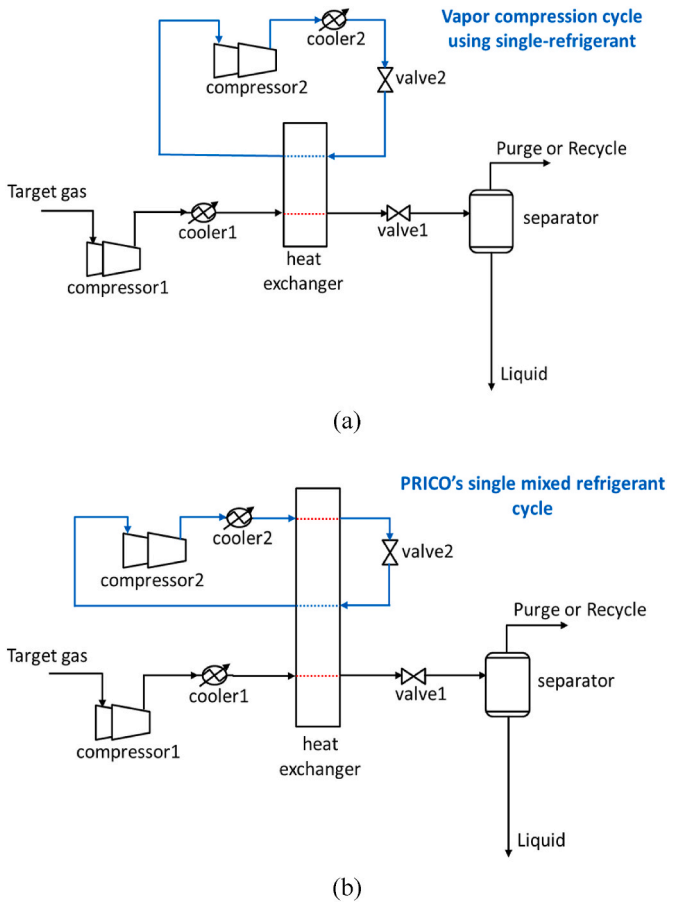


Fig. 1. Diagram of liquefaction systems for (a) Vapor compression cycle using single-refrigerant and (b) PRICO's SMR cycle.

the accumulation of impurities in the process [10].

Fig. 1(b) shows PRICO's SMR process, which has typically been used to liquefy NG or LNG BOG using mixed refrigerants [28–35]. The heat exchange efficiency is increased by using a mixed refrigerant, as the latent heat of each refrigerant in the mixed refrigerants can be utilized for heat exchange. Previous studies have shown that power consumption can be reduced by 30 % when using mixed refrigerants instead of a single refrigerant for LNG BOG re-liquefaction [34]. This indicates that liquefaction efficiency can be enhanced by using mixed-refrigerants when liquefying CO_2 and N_2 mixtures.

3. Process simulation and optimization

3.1. Carbon dioxide storage condition

Table 2 presents the specifications used in this study. Two different filling ratios were considered for the LCO_2 storage tank according to the voyage conditions: laden and ballast voyages. A laden voyage is a voyage from a loading port to an unloading port, while a ballast voyage is a voyage from an unloading port to a loading port. For laden voyages, the filling ratio of the storage tanks was set to 98 % in accordance with the International Code for the Construction and Equipment of Ships Carrying Liquefied Gas (IGC Code) [36]. For ballast voyages, the filling ratio of the storage tanks was set to 5 %, an amount to maintain the temperature of the storage tanks during the ballast voyage and to cool the tanks before loading operation [37,38].

Table 2
LCO_2 storage tank condition.

Parameter	Value
Storage pressure	6.5 bar (Case 1), 15 bar (Case 2)
Storage tank capacity	5000 m^3 × 10 EA
Density	Depends on storage pressure
Filling ratio	Laden voyage: 98 %, Ballast voyage: 5 %
Boil-off rate	0.1 %/day (mass-based)
BOG temperature	−40 °C (Case 1), −20 °C (Case 2)
N_2 on liquid CO_2	0.05–0.15 mol%

The storage pressures were set to 6.5 bar (Case 1) and 15 bar (Case 2), which are commonly mentioned in previous studies (Table 1). At a storage pressure of 6.5 bar, the higher density compared to 15 bar, allows a larger quantity of LCO_2 to be transported for the same volume. However, at 15 bar, the higher feed pressure and lower liquefaction temperature, relative to 6.5 bar, reduce the compression energy consumption required for the re-liquefaction process. The maximum capacity of the storage tank was 5000 m^3 under both 6.5 bar and 15 bar storage pressure conditions [11,12,17]. This study assumed an LCO_2 transport ship with a total of 10 storage tanks, providing a total capacity of 50,000 m^3.

To evaluate the performance of the re-liquefaction process in the presence of impurities in CO_2 BOG, we considered the presence of 0.05–0.15 mol% N_2 impurity in the storage tank. A study on amine-based post combustion carbon capture in a cement plant reported that the captured CO_2 contained impurities of 0.11 mol% N_2 and 0.03 mol% O_2 [10,39]. Among these, N_2 has a lower saturation temperature than O_2 and thus vaporizes more quickly, resulting in more CO_2 BOG and a greater adverse effect on the re-liquefaction process [26]. Therefore, this study focused on N_2 as the impurity and assessed its impact on the re-liquefaction system.

In the storage pressure range of 6.0–19.7 bar, previous studies have reported a boil-off rate (BOR) ranging 0.1–0.15 %/day [16,17,19,40,41] with different insulation structure. In this study, it is assumed that the insulation structure is designed to satisfy BOR of 0.1 % per day, based on studies of Lu et al. [19] and Lee J et al. [17]. The temperature of the CO_2 BOG entering the re-liquefaction process was assumed to be −40 °C at 6.5 bar and −20 °C at 15 bar. This is because in liquefied storage tanks, heated gas collects at the top due to external heat input and exists as superheated gas.

3.2. Composition of carbon dioxide BOG

The filling ratio refers to the volume ratio of LCO_2 space in the storage tank and is calculated based on the volume ratio of liquid to vapor, as shown in equation (1).

$$\text{Filling ratio } [\%] = \frac{V_{liq}}{V_{vap} + V_{liq}} \times 100 = \frac{v_{liq} \bullet (1 - x)}{v_{vap} \bullet x + v_{liq} \bullet (1 - x)} \times 100 \quad (1)$$

Here, V_{vap}, V_{liq} are the volumes of vapor and liquid in the tank and v_{vap}, v_{liq} are the molar volumes of vapor and liquid and x is the vapor mole fraction. In this study, the Peng-Robinson equation of state and the software Aspen HYSYS V14 were used to calculate the vapor mole fraction as a function of the volume ratio of liquid to vapor and extrapolate the resulting vapor composition to the BOG composition. This is because the Peng-Robinson equation of state has shown good predictive performance when compared with experimental vapor-liquid equilibrium data for CO_2/N_2 mixture [42,43].

If an easily evaporating impurity, such as N_2, is present in the LCO_2, the composition of the BOG at equilibrium will change depending on the filling ratio of the storage tank [26]. Since the total volume of the storage tank is fixed at 5000 m^3, a higher filling ratio indicates less space occupied by the CO_2 BOG. The smaller the space occupied by the CO_2 BOG in the storage tank, the greater the molar fraction of N_2 in the BOG that vaporizes faster than CO_2 [26].

Fig. 2 shows the composition of N_2 in BOG as the N_2 composition in LCO_2 varies from 0.05 to 0.15 mol%. At a storage pressure of 6.5 bar, the N_2 composition in the BOG of a laden voyage varied from 4.19 to 12.79

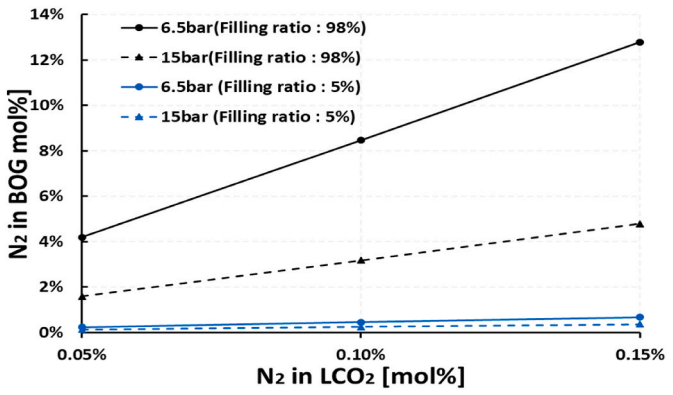


Fig. 2. Variation in the molar fraction of nitrogen in CO_2 BOG as the molar fraction of nitrogen in LCO_2 changes from 0.05 to 0.15 mol%.

mol%, while the N_2 composition in the BOG of a ballast voyage varied from 0.22 to 0.66 mol%. At a storage pressure of 15 bar, the N_2 composition in the BOG of a laden and ballast voyage varied 1.58–4.79 mol% and 0.12–0.35 mol%, respectively.

3.3. Carbon dioxide BOG re-liquefaction process

Fig. 3 shows the PFDs used in this study. Fig. 3(a) shows the re-liquefaction process with a single refrigerant, ammonia, as a conventional re-liquefaction process. The CO_2 BOG was compressed to high pressure through two-stage compression and cooled to a low temperature through heat exchange with ammonia in a heat exchanger. It was then partially liquefied by JT expansion to the storage pressure through a JT valve and separated into gas and liquid in a separator. The liquid was sent to the storage tank, and the vapor was recycled to the compressor. If the N_2 content was high, some vapor needed to be purged to prevent accumulation in the cycle. Fig. 3(b) shows an alternative process using ammonia refrigerant, where an additional separator was installed after the heat exchanger to improve the separation efficiency. The first separator primarily purged N_2, which vaporizes more easily than CO_2, and the relatively pure CO_2 was then sent to the final expansion valve.

Fig. 3(c) shows the proposed process using a mixed refrigerant with two-stage separators. The mixed refrigerant consists of ethane (C2), propane (C3), normal butane (nC4), and iso-butane (iC4). For pure CO_2, the dew point was −50.3 °C at a storage pressure of 6.5 bar, and the dew point decreased with higher N_2 impurity. To achieve a lower cooling temperature, propane, with a boiling point of −42.1 °C at atmospheric pressure, was added to ethane, which has a boiling point of −88.5 °C at atmospheric pressure forming the mixed refrigerant. Additionally, to improve heat exchange efficiency in the high-temperature section of the heat exchanger, n- butane and iso-butane are included, with boiling points of approximately 0.5 °C and −11.7 °C at atmospheric pressure, respectively. Table 3 shows the process conditions used in the process simulations.

3.4. Process optimization

To optimize the performance of each re-liquefaction process, the annual CO_2 emissions from the re-liquefaction process were defined as an objective function and minimized. The annual CO_2 emissions from the re-liquefaction process consisted of the CO_2 generated by ship's auxiliary generator to provide compression power and the CO_2 purged during the re-liquefaction process operation. By considering the annual CO_2 emissions, both the environmental impact and energy consumption of the process were assessed simultaneously (see Fig. 4).

Annual CO_2 emissions ($m_{\text{ann CO}_2}$) were determined as the sum of CO_2 emissions during ballast voyages ($m_{\text{Ballast CO}_2}$) and CO_2 emissions during laden voyages ($m_{\text{Laden CO}_2}$) as shown in equations (2)–(4).

$$m_{\text{ann CO}_2} = m_{\text{Laden CO}_2} + m_{\text{Ballast CO}_2} \tag{2}$$

$$m_{\text{Laden CO}_2} = \dot{m}_{\text{Laden purge}} \bullet t_{\text{Laden}} + \dot{m}_{\text{Laden power}} \bullet t_{\text{Laden}} \tag{3}$$

$$m_{\text{Ballast CO}_2} = \dot{m}_{\text{Ballast purge}} \bullet t_{\text{Ballast}} + \dot{m}_{\text{Ballast power}} \bullet t_{\text{Ballast}} \tag{4}$$

Here, $\dot{m}_{\text{Laden purge}}$ and $\dot{m}_{\text{Ballast purge}}$ represent the amount of CO_2 purged per hour from the re-liquefaction process, t_{Laden} and t_{Ballast} were assumed to be 4000 h per year of operation of the re-liquefaction facility during the laden and ballast voyages, respectively.

$\dot{m}_{\text{Laden power}}$ and $\dot{m}_{\text{Ballast power}}$ represent the amount of CO_2 per hour generated by the compression power consumption for the re-liquefaction during the laden and ballast voyages, as shown in equations (5) and (6).

$$\dot{m}_{\text{Ballast power}} = \dot{W}_{\text{Ballast}} \bullet \text{SFOC} \cdot C_f \tag{5}$$

$$\dot{m}_{\text{Laden power}} = \dot{W}_{\text{Laden}} \bullet \text{SFOC} \cdot C_f \tag{6}$$

Here, $\dot{W}_{\text{Ballast}}$ and $\dot{W}_{\text{Laden}}$ represent the compression power consumption of the BOG re-liquefaction system during ballast and laden voyages, respectively. SFOC is the amount of fuel (MGO) consumed by the generator per kWh, assumed to be 197.5 g_{fuel}/kWh [44]. C_f is the amount of CO_2 generated by using one ton of MGO, defined as 3.206 $\text{ton}_{\text{CO2}}/\text{ton}_{\text{fuel}}$ by the International Maritime Organization (IMO) [45,46].

To minimize annual CO_2 emissions in the re-liquefaction process, the refrigerant temperature after expansion, compression pressures, refrigerant composition and purge ratio were defined as process variables as shown in Table 4 and Fig. 3. Process optimization was performed using the particle swarm optimization (PSO) algorithm, which is an effective method for solving nonlinear and multidimensional problems like liquefaction processes and has been applied in previous studies on liquefaction process optimization [17,32,34].

In optimizing the re-liquefaction process, process constraints were set, as shown in Table 4. The vapor fraction at the compressor inlet was defined as a constraint to prevent liquid from entering the compressor. To avoid dry ice formation, constraints were imposed to ensure that CO_2 remained above −55 °C even in the lowest temperature intervals (S8) in the process after JT expansion. If this constraint was violated, a penalty value was assigned to prevent the selection of the optimal value.

3.5. Cost estimation method

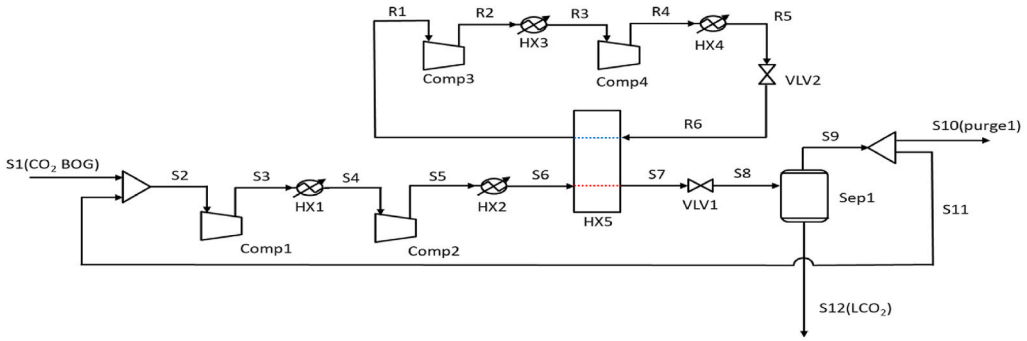
The capital expenditure (CAPEX) was calculated based on the equipment capacity determined by each simulation, and included equipment purchase and installation costs. The calculation of the CAPEX was performed using the Aspen Process Economic Analyzer V14 (APEA V14) according to Table 5. The final CAPEX was calculated as the annualized CAPEX considering an operating period (L) of 20 years and an interest rate (i) of 10 % [17]. This is shown in equation (7).

$$\text{Annualized CAPEX} = \text{CAPEX} \bullet \frac{i \bullet (1+i)^L}{(1+i)^L - 1} \tag{7}$$

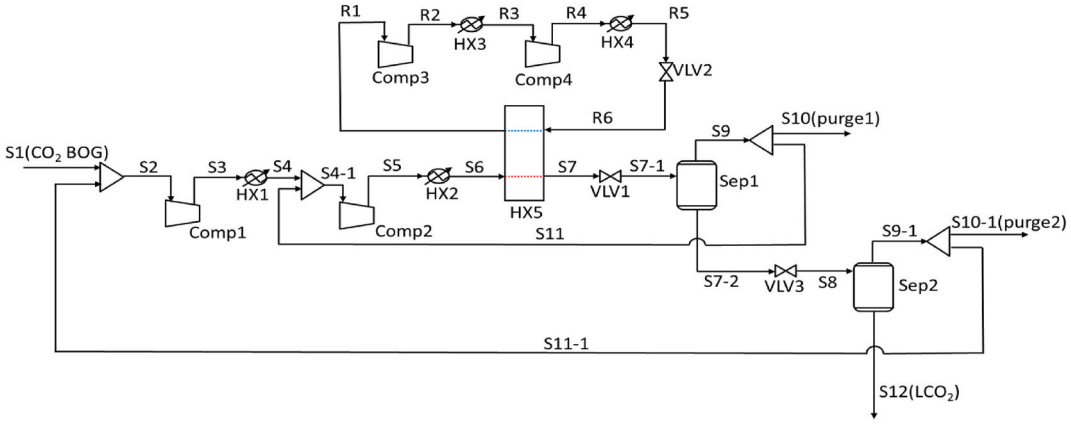
The operating expenditure (OPEX) considered only the compression power consumption in the re-liquefaction process, and was estimated as the cost of the fuel required to generate power in the ship's generator. It was calculated by multiplying compression power ($\dot{W}$), SFOC, and fuel price (Cos t_{fuel}), as shown in equation (8). The fuel used was marine gas oil (MGO) and the fuel price was assumed 825 USD/ton_{fuel} [47].

$$\text{OPEX} = \dot{W} \bullet \text{SFOC} \bullet \text{Cos } t_{\text{fuel}} \tag{8}$$

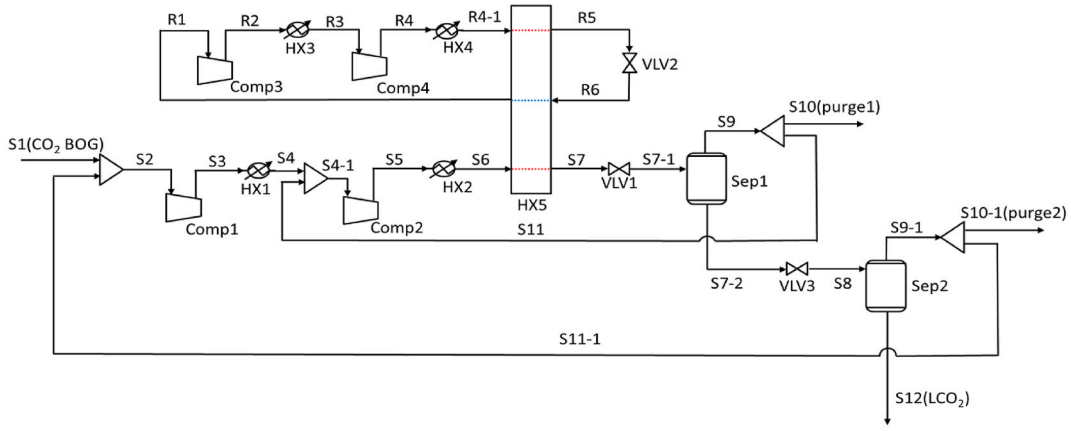
The total annualized cost (TAC) of each re-liquefaction process was calculated based on the optimization results. TAC was defined as the



(a) Case 1-1 & Case 2-1



(b) Case 1-2 & Case 2-2



(c) Case 1-3 & Case 2-3

Fig. 3. PFD of the CO_2 BOG re-liquefaction process for (a) conventional single refrigerant cycle (Case 1-1 and Case 2-1), (b) single refrigerant cycle with two-stage separation (Case 1–2 and Case 2-2), (c) mixed refrigerant cycle with two-stage separation (Case 1–3 and Case 2–3).

Table 3
Process specifications of the CO_2 BOG re-liquefaction system.

Items	Contents
Equation of state	Peng-Robinson [10,16,20,22,23]
Storage pressure	Case 1-1, 1–2, 1–3: 6.5 bar Case 2-1, 2-2, 2–3: 15 bar
Refrigerant	Case 1-1, 1–2, 2-1, 2-2: NH3 Case 1–3, 2–3: C2, C3, nC4, iC4
Compressor adiabatic efficiency	75 %
Maximum pressure ratio of compressor	3.0
Cooling water temperature	30 °C
Cooler minimum temp. approach	5.0 °C
Heat exchanger minimum temp. approach	3.0 °C

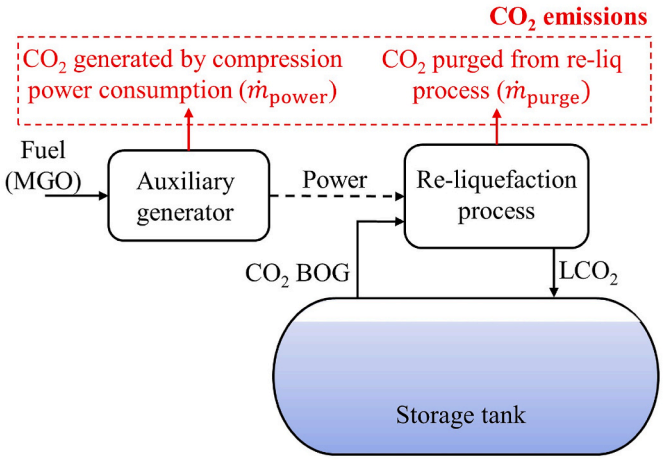


Fig. 4. Schematic diagram of the CO_2 re-liquefaction process with auxiliary generator for evaluating annual CO_2 emission.

Table 4
Process constraints and variables for optimization.

	Variable	Remark
Process Variable	C2, C3, nC4, iC4 mass flow (ton/h)	R1(Case 1–3, 2–3)
	Pressure ratio	Comp 1,2,3
	Pressure (bar)	R6 R5(Case 1–3, 2–3)
	Temperature (°C)	R1 (Case 1-1, 1–2, 2-1, 2-2) R5 (Case 1–3, Case 2–3)
	Purge ratio	Purge1 ratio Purge2 ratio (Case 1–2, 2-2, 1–3, 2–3)
Process Constraints	Vapor fraction = 1.0	Comp 1, 2, 3, 4 inlet
	Minimum temp. approach ≥3.0 (°C)	HX5
	Pressure ratio of compressor ≤3.0	Comp 1, 2, 3, 4
	CO_2 temperature ≥ -55.0 (°C)	S8

Table 5
CAPEX estimation method.

Equipment	Mapping	Estimation Method	Base year
Compressor	Centrifugal	Aspen Process Economic Analyzer (APEA) V14.	2022
Heat Exchanger	Shell & Tube		
Separator	Pressurized Vessel		
Storage Tank	Pressurized Vessel		

sum of annualized capital expenditure (CAPEX) and operating expenditure (OPEX), as shown in equation (9).

$$TAC = \text{Annualized CAPEX} + OPEX \tag{9}$$

When the re-liquefaction process emitted CO_2, the economic loss due to carbon tax needed to be considered. Therefore, the total annual expenditure (TAE) of the re-liquefaction process was calculated as the sum of TAC and the annual carbon tax as shown in equation (10). The carbon tax per ton of CO_2 emissions (Tax_{CO_2}) was assumed to be 150 USD/ton_{CO2} [48], and the annual carbon tax ($Tax_{ann\ CO_2}$) was calculated using equations (10) and (11).

$$TAE = TAC + Tax_{ann\ CO_2} \tag{10}$$

$$Tax_{ann\ CO_2} = Tax_{CO_2} \bullet m_{ann\ CO_2} \tag{11}$$

Additionally, purge in the re-liquefaction process leads to CO_2 cargo loss, and the density of LCO_2 varies with storage pressure, so the annual transported CO_2 amount must be considered in the economic evaluation. To calculate the annual transported CO_2, it was assumed that the CO_2 carrier made a total 6 voyages per year. The annual transported CO_2 amount ($m_{ann\ transport}$) is calculated by multiplying the amount of cargo unloaded per voyage (m_{cargo}) by the annual number of voyages (N_{Voyage}), then subtracting the purge amounts during laden voyages ($\dot{m}_{Laden\ purge} \bullet t_{Laden}$) and ballast voyages ($\dot{m}_{Ballast\ purge} \bullet t_{Ballast}$). This is shown in equation (12).

$$m_{ann\ transport} = m_{cargo} \bullet N_{Voyage} - \left(\dot{m}_{Laden\ purge} \bullet t_{Laden} + \dot{m}_{Ballast\ purge} \bullet t_{Ballast}\right) \tag{12}$$

This study proposed an index of cost per transportation (CPT) as shown in equation (13), considering the total annual expenditure of the re-liquefaction process and the annual transported CO_2 amount. CPT could determine the most economical re-liquefaction process and storage pressure.

$$CPT = \frac{TAC + Tax_{ann\ CO2}}{m_{ann\ transport}} = \frac{TAE}{m_{ann\ transport}} \tag{13}$$

4. Results and discussion

4.1. CO_2 emissions of re-liquefaction process

The annual CO_2 emissions for each case are summarized in Table 6, which shows that most CO_2 emissions occur during the laden voyage. CO_2 emissions during the ballast voyage accounted for only 0.6–3.1 % of CO_2 emissions during the laden voyage. This is because the storage tank filling ratio during the ballast voyage was 5 %, significantly lower than the 98 % during the laden voyage. As a result, the amount of BOG generated during the ballast voyage was lower, leading to a reduction in compression power consumption and purge. Furthermore, according to Fig. 2, the N_2 content in the BOG during the ballast voyage was a maximum of 0.66 mol%, significantly lower than the maximum of 12.79 mol% during the laden voyage. Consequently, the proportion of non-condensable gases that be purged during the ballast voyage was reduced compared to the laden voyage. For example, during the ballast voyage, the purge ratio for Case 1–1 ranges from 0.0 to 0.07. In contrast, during the laden voyage, the purge ratio rises significantly to 0.73–0.99 (Table A. 1 and Table A. 2). As a result, the annual CO_2 emissions from the re-liquefaction process were predominantly determined by the CO_2 emissions during the laden voyage. Therefore, the performance of each CO_2 BOG re-liquefaction process was compared based on the CO_2 emissions during the laden voyage.

Fig. 5 shows the annual CO_2 emissions from purge and compression power generation during the laden voyage for each case. Comparing the conventional processes of Case 1-1 (storage pressure 6.5 bar) and Case 2-1 (storage pressure 15 bar), Case 1-1 exhibited significantly higher emissions, particularly from purge. The CO_2 emissions from purge in

Table 6
The annual CO_2 emissions of the re-liquefaction process.

N_2 in LCO_2	0.05 [mol%]			0.1 [mol%]			0.15 [mol%]		
	$m_{Laden\ CO_2}$ [ton_{CO2}]	$m_{Ballast\ CO_2}$ [ton_{CO2}]	$m_{ann\ CO_2}$ [ton_{CO2}]	$m_{Laden\ CO_2}$ [ton_{CO2}]	$m_{Ballast\ CO_2}$ [ton_{CO2}]	$m_{ann\ CO_2}$ [ton_{CO2}]	$m_{Laden\ CO_2}$ [ton_{CO2}]	$m_{Ballast\ CO_2}$ [ton_{CO2}]	$m_{ann\ CO_2}$ [ton_{CO2}]
Case 1-1	2385.1	27.3	2412.4	4477.5	32.7	4510.2	6717.7	37.8	6755.5
Case 1-2	1042.6	26.3	1068.9	1516.3	29.4	1545.7	1973.6	31.7	2005.3
Case 1-3	916.2	24.8	941.0	1392.8	25.8	1418.6	1891.1	27.3	1918.4
Case 2-1	467.2	12.0	479.2	662.3	12.9	675.2	804.9	13.8	818.7
Case 2-2	423.1	11.9	435.0	553.6	12.6	566.2	669.8	13.7	683.5
Case 2-3	365.5	11.5	377.0	442.7	12.4	455.1	513.3	13.7	527.0

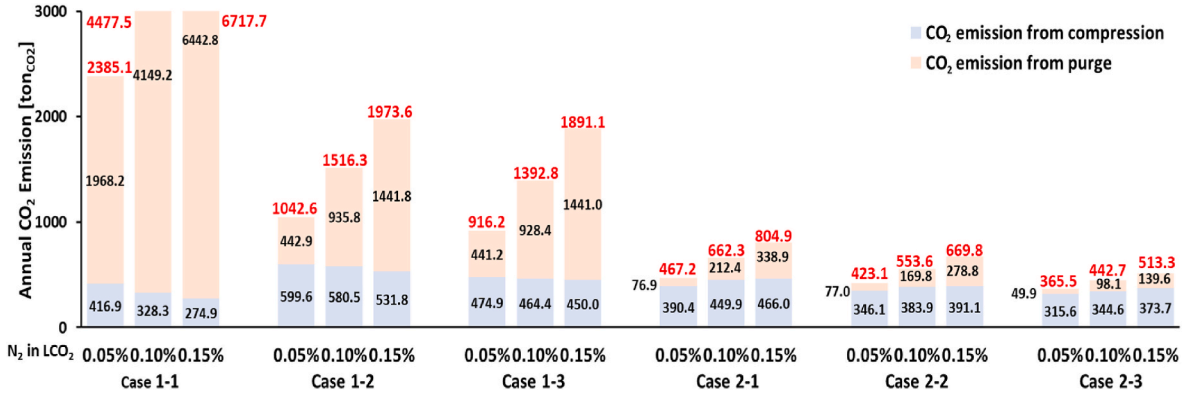


Fig. 5. CO_2 emissions from purge and compression power during the laden voyage.

Table 7
Results of the main streams of each CO_2 BOG re-liquefaction processes at the laden voyage.

N_2 in LCO_2		0.05 (mol%)			0.10 (mol%)			0.15 (mol%)		
Stream		S1	S7	S8	S1	S7	S8	S1	S7	S8
Case 1-1	P (bar)	6.5	47.1	6.5	6.5	37.7	6.5	6.5	39.8	6.5
	T (°C)	−40	−13.7	−55	−40	−9.4	−55	−40	−5.5	−55
	Vap. Frac.	1.0	0.02	0.31	1.0	0.34	0.54	1.0	0.66	0.77
	Purge ratio1	0.73			0.89			0.99		
Case 1-2	P (bar)	6.5	58.4	6.5	6.5	57.4	6.5	6.5	58.5	6.5
	T (°C)	−40	−21.8	−55	−40	−20.6	−55	−40	−21.9	−55
	Vap. Frac.	1.0	0.24	0.09	1.0	0.31	0.09	1.0	0.24	0.09
	Purge ratio1	0.14			0.25			0.54		
	Purge ratio2	0.0			0.0			0.0		
Case 1-3	P (bar)	6.5	30.6	6.5	6.5	29.4	6.5	6.5	30.6	6.5
	T (°C)	−40	−34.2	−55	−40	−34.3	−55	−40	−32.8	−55
	Vap. Frac.	1.0	0.17	0.09	1.0	0.22	0.09	1.0	0.31	0.09
	Purge ratio1	0.29			0.54			0.62		
	Purge ratio2	0.0			0.0			0.0		
Case 2-1	P (bar)	15	48.8	15	15	64.3	15	15	69.7	15
	T (°C)	−20	−22.0	−43.1	−20	−22.0	−46.7	−20	−22.0	−48.4
	Vap. Frac.	1.0	0.15	0.31	1.0	0.19	0.38	1.0	0.24	0.43
	Purge ratio1	0.03			0.08			0.11		
Case 2-2	P (bar)	15	45.0	15	15	68.8	15	15	70.0	15
	T (°C)	−20	−22.0	−38.2	−20	−22.0	−40.4	−20	−22.0	−40.8
	Vap. Frac.	1.0	0.07	0.15	1.0	0.10	0.14	1.0	0.18	0.14
	Purge ratio1	0.2			0.16			0.19		
	Purge ratio2	0.0			0.0			0.0		
Case 2-3	P (bar)	15	30.0	15	15	38.6	15	15	44.3	15
	T (°C)	−20	−36.4	−42.3	−20	−38.6	−45.7	−20	−40.8	−48.2
	Vap. Frac.	1.0	0.18	0.06	1.0	0.09	0.09	1.0	0.14	0.1
	Purge ratio1	0.04			0.3			0.3		
	Purge ratio2	0.04			0.0			0.0		

Case 2-1 ranged from 76.9 to 338.9 tons, whereas in Case 1-1, they ranged from 1968.2 to 6442.8 tons. This difference resulted from the temperature constraint, which was set to prevent the CO_2 temperature from falling below −55 °C to avoid dry ice formation in the re-liquefaction process. Table 7 presents the simulation results for the main streams (S1, S7, and S8) of each re-liquefaction process during the laden voyage, as shown in Fig. 3. In Case 1-1, the temperatures of stream S8 after JT expansion were all −55 °C, and the vapor fraction of S8 was very high (0.31–0.77), resulting in, a high purge ratio of 0.73–0.99. As the N_2 content increased, the dew point of the CO_2 mixture became lower, requiring a lower temperature for liquefaction. However, the temperature constraint of −55 °C prevented most of the CO_2 in Case 1-1

from being liquefied, necessitating its purge. In Case 2-1, the temperature of stream S8 ranged from −43.1 to −48.4 °C, indicating that Case 2-1 was not affected by the temperature constraint. This was because the CO_2 mixture in Case 2-1 had a higher dew point temperature due to the higher storage pressure compared to Case 1-1.

Fig. 5 also shows that the two-stage vapor-liquid separation in Cases 1–2, 1–3, 2-2, and 2–3 reduced CO_2 emissions from purge compared to the conventional re-liquefaction processes in Cases 1-1 and 2-1. Cases 1–2 and 1–3 exhibited approximately 22.5 % lower CO_2 emissions from purge compared to Case 1-1. This reduction occurred because the first high pressure separator facilitated the pre-separation of N_2, which is lighter than CO_2. Consequently, under the temperature constraint of −55 °C, stream S8 achieved a vapor fraction as low as 0.09, as the CO_2, with higher purity, expanded to the tank storage pressure (Table 7). Similarly, Cases 2-2 and 2–3 also achieved up to 41.1 % lower CO_2 emissions from purge compared to Case 2-1.

Additionally, the application of a mixed refrigerant further reduced CO_2 emissions from compression power, as shown in Cases 1–3 and 2–3 in Fig. 5. Case 1–3 exhibited approximately 79.2–84.6 % lower CO_2 emissions from compression power compared to Case 1–2. This reduction occurred because the use of a mixed refrigerant decreased the maximum CO_2 pressure. Table 7 shows that the maximum CO_2 pressure in stream S7 was 58.4–58.5 bar in Case 1–2 but decreased to 29.4–30.6 bar in Case 1–3. Consequently, Case 1–3 required 177.7–187.5 kW of compression power, compared to 210.0–236.7 kW in Case 1–2, as illustrated in Fig. 6. Similarly, Case 2–3 demonstrated approximately 89.8–95.6 % lower CO_2 emissions from compression power than Case 2-2, due to the reduced compression pressure required in Case 2–3.

The improvement in energy efficiency with the use of mixed refrigerants is more clearly illustrated through the heat composite curve of the heat exchanger (HX5). As shown in Fig. 7, the temperature difference between the heat composite curves in HX5 decreased from a maximum of 43.5 °C and an average of 24.8 °C in Case 1–2 to a maximum of 29.8 °C and an average of 11.6 °C in Case 1–3. This reduction occurred because single refrigerants evaporate at a constant temperature, whereas mixed refrigerants evaporate at varying temperatures for each component. This brought the hot composite curve and cold composite curve closer together, enabling more uniform heat transfer. Consequently, Case 1–3 utilized cold energy more efficiently and achieved significantly higher energy efficiency compared to Case 1–2. Similarly, the temperature difference between the heat composite curves in HX5 decreased from a maximum of 46.3 °C and an average of 28.1 °C in Case 2-2 to a maximum of 27.1 °C and an average of 13.7 °C in Case 2–3. These results indicate that Case 2–3 also achieved better energy efficiency than Case 2-2.

4.2. Cost estimation of CO_2 BOG re-liquefaction process

As shown in Fig. 8 and Table A. 6, the TAC was more economical in Case 1 (storage pressure 6.5 bar) compared to Case 2 (storage pressure 15 bar). This was primarily due to the cost of storage tanks, which accounted for more than 70 % of the TAC. The TAC for the storage tank in Case 2 was 0.68 million USD per year higher than that in Case 1, due to the higher storage pressure.

However, when considering the $Tax_{ann\ CO_2}$, the TAE of Case 1–1 may be worse than that of Case 2. This was because Case 1-1 had the largest increase in annual CO_2 emissions. As shown in Fig. 8, when the N_2 content in LCO_2 increased from 0.05 to 0.15 mol%, the $Tax_{ann\ CO_2}$ for Case 1-1 increased by 0.36–1.01 million USD/year. Therefore, with 0.15 mol% N_2 in LCO_2, the TAE of Case 1-1 was higher than that of Case 2 by 0.09–0.23 million USD.

Case 1–2 and Case 1–3 demonstrate that applying two-stage gas-liquid separation and mixed refrigerants achieved significant economic benefits. First, applying two stage separation reduced CO_2 emissions from purge resulting in a lower $Tax_{ann\ CO_2}$. As shown in Fig. 8, Case 1–2 had a TAE that was 0.09–0.55 million USD/year lower than Case 1-1. Second, the use of mixed refrigerants further reduced compression power, leading to lower OPEX and BOG compressor costs. Case 1–3 had a TAE that is 0.2–0.64 million USD/year lower than Case 1-1.

The economic evaluation results of CPT are shown in Fig. 9. In most cases, the CPTs of Case 1 were lower than those of Case 2. This was due to the CO_2 density in Case 1 being approximately 8.5 % higher than in Case 2, allowing for more cargo to be transported within the same volume. In Case 1–1, N_2 content increased, CPT increased rapidly due to cargo loss from CO_2 purging. However, Cases 1–2 and 1–3 reduced cargo loss from purging by applying two-stage separation and mixed refrigerants, showing a cost advantage of 2.16–2.41 USD/ton_{CO2} compared to Cases 2-2 and 2–3. As a result, Case 1–3 was the most economical case with the lowest CPT of 8.9–9.3 USD/ton_{CO2}.

The varying carbon tax (Tax_{CO_2}) significantly affected CPT. In general, venting CO_2 BOG while paying carbon tax is less economical than the re-liquefaction process (Fig.B. 1). Fig. 10 shows the change in re-liquefaction process's CPT with carbon tax at 0.1 mol% N_2, with Case 1-1 being highly dependent on the carbon tax. When the carbon tax exceeded 270 USD/ton_{CO2}, Case 1-1 had the highest CPT than other cases. However, Case 1–2 and 1–3 remained more economical than Case 2, even if the carbon tax increased up to 300 USD/ton_{CO2}.

5. Conclusion

This study proposed new two-stage separation re-liquefaction processes using mixed refrigerants for CO_2 BOG, considering N_2 impurity under different storage conditions of 6.5 bar (Case 1) and 15 bar (Case 2). The proposed processes, which apply two-stage separation (Case 1–2,

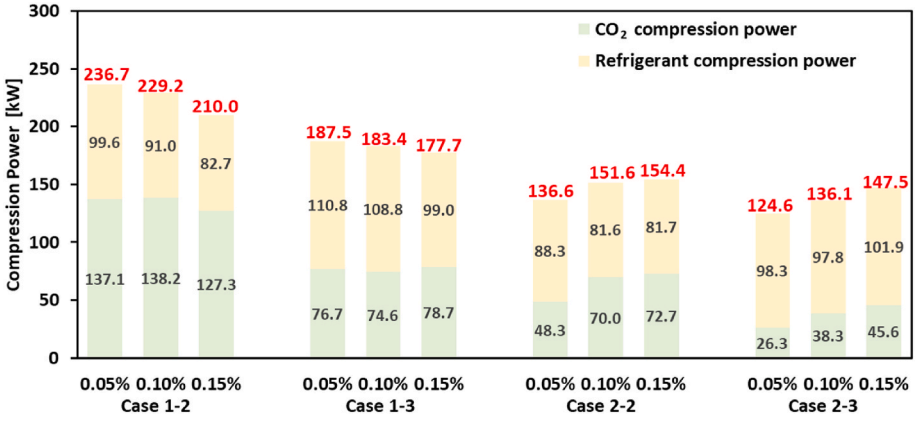


Fig. 6. Compression power during laden voyage.

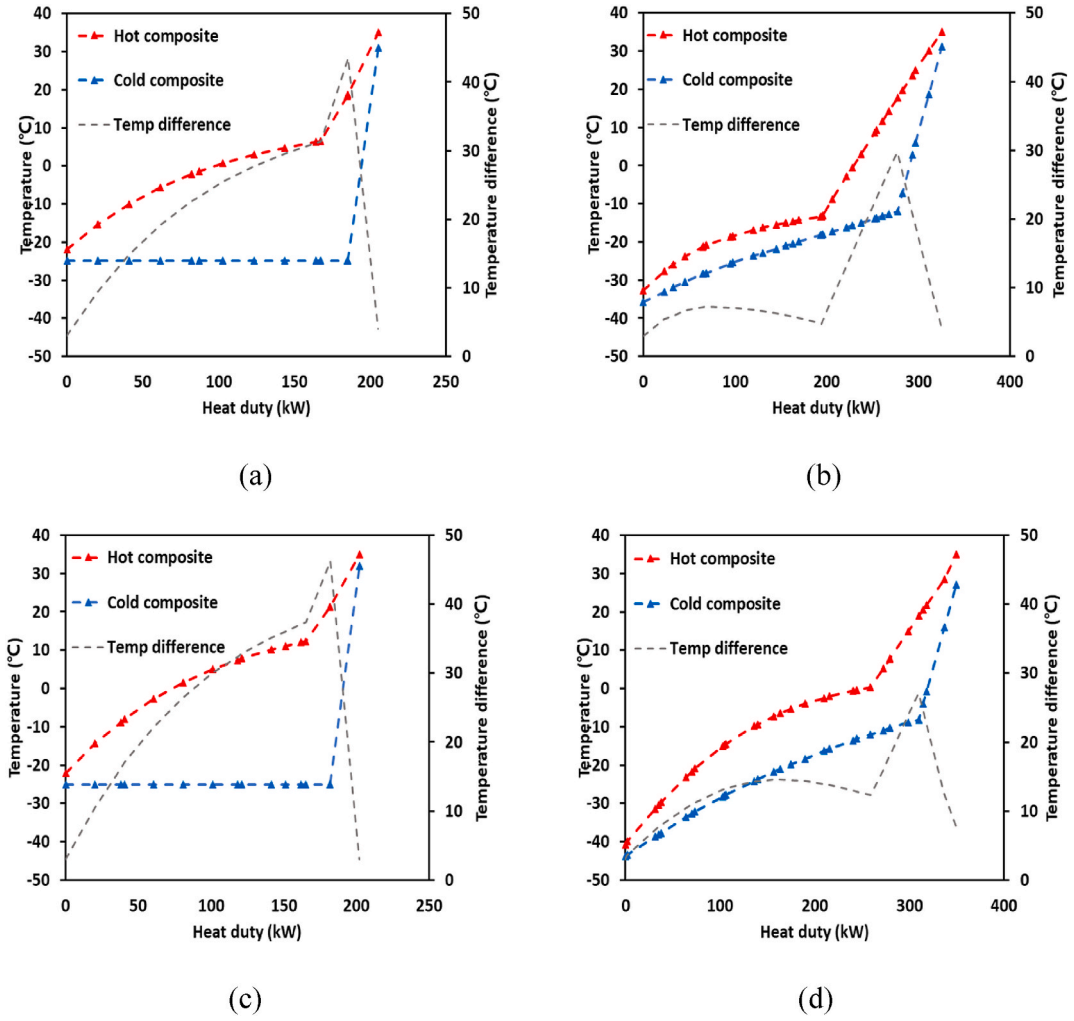


Fig. 7. Heat composite curve for (a) Case 1–2, (b) Case 1–3, (c) Case 2-2, and (d) Case 2–3 at 0.15 mol% N_2, laden voyage.

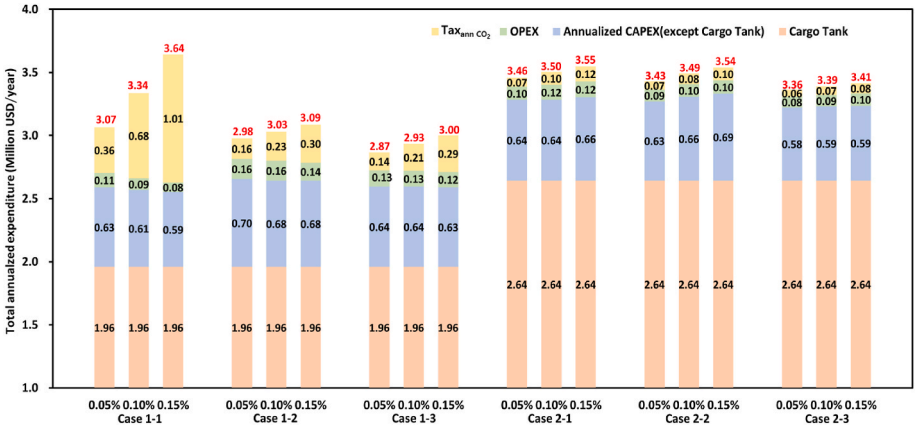


Fig. 8. Results of TAE (Total annualized expenditure).

2-2) and both two-stage separation and mixed-refrigerants (Case 1–3, 2–3), were analyzed. The performance of the proposed processes was compared to conventional processes (Case 1-1, 2-1) based on annual CO_2 emissions, considering not only the energy consumption of the process but also its indirect impact on GHG emissions.

The simulation results showed that the proposed two-stage separation process effectively reduced annual CO_2 emissions compared to the conventional process by decreasing purge amounts. Additionally, the two-stage separation process with mixed refrigerants further reduced the annual CO_2 emissions by decreasing energy consumption. Specifically, at 6.5 bar, compared to Case 1-1, Case 1–2 and Case 1–3 emitted only 29.7–44.3 % and 28.4–39.0 % of the annual CO_2 emissions, respectively. At 15 bar, compared to Case 2-1, Case 2-2 and 2–3 emitted only 83.5–90.8 % and 64.4–78.7 % of the annual CO_2 emissions,

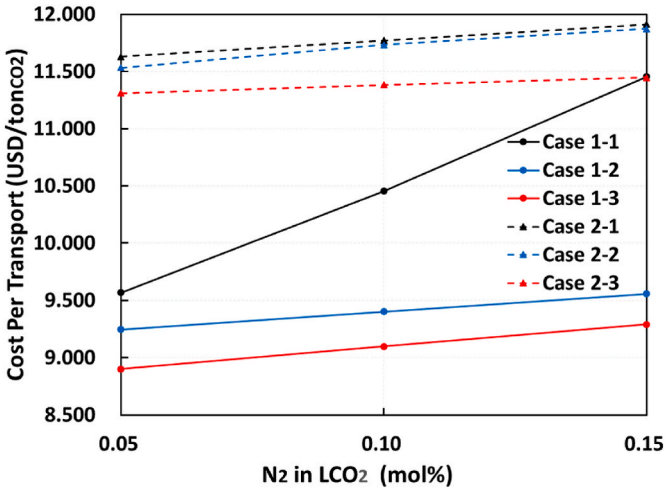


Fig. 9. Results of CPT (Cost per transport).

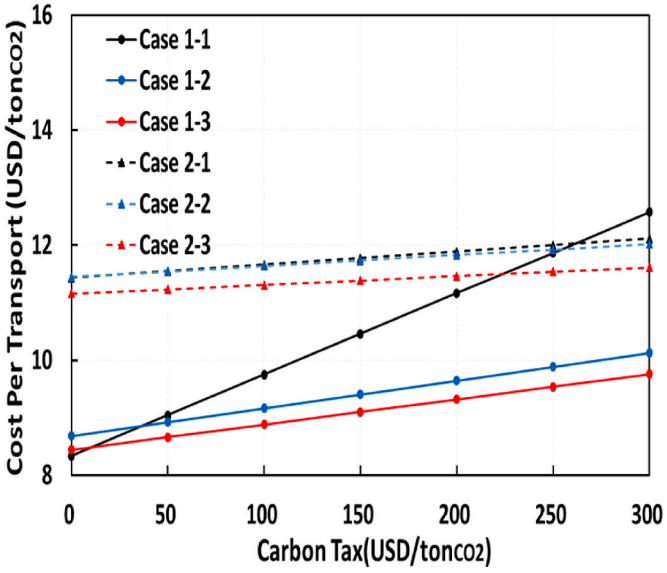


Fig. 10. Variation in carbon tax from 0 USD to 300 USD at N_2 0.1 mol%.

respectively. At 15 bar, the proposed cases achieved lower annual CO_2 emissions because they required less energy for liquefaction due to their relatively higher liquefaction temperature. As a result, Case 2–3 showed the lowest annual CO_2 emissions among all cases, ranging from 377.0 to 527.0 tons.

The economic evaluation, considering annual CO_2 emissions and the annual transported CO_2 amount, showed that the CPTs at 6.5 bar (Case 1–2 and 1–3) ranged from 8.90 to 9.56 USD/ton_{CO2}, lower than those of Case 2-2 and 2–3, which ranged from 11.31 to 11.87 USD/ton_{CO2}. This was because CO_2 had higher density at the low storage pressure in Case 1, reducing tank costs. In conclusion, Case 1–3 exhibited the lowest CPT among Case 1, ranging from 8.90 to 9.29 USD/ton_{CO2}.

CRediT authorship contribution statement

Ijun Jeong: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Yoonju Jeon:** Investigation, Conceptualization. **Hongkyoung Shin:** Validation, Methodology. **Jaejun Lee:** Validation, Conceptualization. **Youngsub Lim:** Supervision, Resources, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Nomenclature

Abbreviation

CO_2	Carbon dioxide
N_2	Nitrogen
GHG	Greenhouse gas
CCS	Carbon capture and storage
BOG	Boil-off gas
CHS	Cargo handling system
LCO_2	Liquid carbon dioxide
NG	Natural gas
LNG	Liquefied natural gas
TAC	Total annualized cost
TAE	Total annual expenditure
CPT	Cost per transportation
JT	Joule-Thomson
BOR	Boil-off rate
C2	Ethane
C3	Propane
nC4	Normal butane
iC4	Iso-butane
IMO	International Maritime Organization
MGO	Marine gas oil

Symbols

V_{vap}	volume of gas [m^3]
V_{liq}	volume of liquid [m^3]
v_{vap}	molar volume of vapor [m^3/kmol]
v_{liq}	molar volume of liquid [m^3/kmol]
x	vapor mole fraction [−]
$m_{ann\ CO_2}$	annual CO_2 emissions [ton_{CO2}]
$m_{Laden\ CO_2}$	CO_2 emissions during laden voyage [ton_{CO2}]
$m_{Ballast\ CO_2}$	CO_2 emissions during ballast voyage [ton_{CO2}]
$\dot{m}_{Laden\ purge}$	CO_2 emission by purge at laden voyage [ton_{CO2}/hour]
$\dot{m}_{Ballast\ purge}$	CO_2 emission by purge at ballast voyage [ton_{CO2}/hour]
$\dot{m}_{Laden\ power}$	CO_2 emission by re-liquefaction power at laden voyage [ton_{CO2}/hour]
$\dot{m}_{Ballast\ power}$	CO_2 emission by re-liquefaction power at ballast voyage [ton_{CO2}/hour]
t_{Laden}	Operating hours of the re-liquefaction system during the annual laden voyage [hour]
$t_{Ballast}$	Operating hours of the re-liquefaction system during the annual laden voyage [hour]
$\dot{W}_{Laden}$	Total compression power for re-liquefaction system at laden voyage [kW]
$\dot{W}_{Ballast}$	Total compression power for re-liquefaction system at ballast voyage [kW]
SFOC	Specific fuel oil consumption [ton_{fuel}/kWh]
Cos t_{fuel}	Cost of fuel [USD/ton_{fuel}]
C_f	Emission factor for MGO [ton_{CO2}/ton_{fuel}]
$m_{ann\ transport}$	Annual transported CO_2 amount [ton_{CO2}]
m_{cargo}	Amount of CO_2 unloaded per voyage [ton_{CO2}/voyage]
N_{Voyage}	Annual number of voyages [voyage]
$Tax_{ann\ CO_2}$	Annual tax due to CO_2 emissions [USD]
Tax_{CO_2}	Tax per ton of CO_2 [USD/ton_{CO2}]

APPENDIX A. Result of optimization and cost estimation

Table A. 1
Optimization results of CO_2 BOG re-liquefaction processes at laden voyage

N_2 in LCO_2 (mol%)		0.05						0.10						0.15					
Case		1–1	1–2	1–3	2–1	2–2	2–3	1–1	1–2	1–3	2–1	2–2	2–3	1–1	1–2	1–3	2–1	2–2	2–3
Pressure ratio	PR_{comp1}	3.0	3.0	3.0	2.78	2.99	2.0	3.0	2.95	3.0	2.98	3.0	2.57	3.0	3.0	3.0	3.0	3.0	2.95
	PR_{comp2}	2.42	3.0	1.57	1.17	1.01	1.01	1.94	3.0	1.51	1.44	1.53	1.01	2.04	3.0	1.57	1.55	1.55	1.01
	PR_{comp3}	2.75	3.0	2.88	2.99	3.0	2.98	2.81	2.92	2.70	3.0	3.0	2.93	2.28	3.0	2.75	3.0	3.0	2.85
Pressure (bar)	P_{R5}	–	–	18.18	–	–	11.92	–	–	18.76	–	–	2.24	–	–	2.37	–	–	16.28
	P_{R6}:	2.17	1.51	2.69	1.49	1.49	1.48	2.60	1.59	2.70	1.49	1.49	16.73	3.05	1.50	17.07	1.49	1.49	1.97
Temperature (°C)	T_{R1}	32	32	–	31.81	32	–	32	32	–	32	32	–	32	31.0	–	32	32	–
	T_{R5}	–	–	−30.3	–	–	−36.4	–	–	−28.2	–	–	−38.6	–	–	−21.28	–	–	−39.9
Purge ratio	Purge1	0.73	0.14	0.29	0.03	0.2	0.04	0.89	0.25	0.54	0.08	0.16	0.3	0.99	0.54	0.62	0.11	0.19	0.30
	Purge2	–	0.0	0.0	–	0.0	0.04	–	0.0	0.0	–	0.0	0.0	–	0.0	0.0	–	0.0	0.0
Refrigerant composition (mol%)	C_{C2}	–	–	26.89	–	–	12.79	–	–	26.56	–	–	26.19	–	–	22.31	–	–	24.66
	C_{C3}	–	–	57.62	–	–	43.70	–	–	59.71	–	–	39.90	–	–	58.61	–	–	42.32
	C_{iC4}	–	–	5.74	–	–	20.91	–	–	6.18	–	–	15.96	–	–	10.94	–	–	10.66
	C_{nC4}	–	–	14.75	–	–	22.59	–	–	7.55	–	–	17.95	–	–	8.14	–	–	22.36
CO_2 emission (ton)		2385.1	1042.6	916.2	467.2	423.1	365.5	4477.5	1516.3	1392.8	662.3	553.6	442.7	6717.7	1973.6	1891.1	804.9	669.8	513.3

Table A. 2
Optimization results of CO_2 BOG re-liquefaction processes at ballast voyage

N_2 in LCO_2 (mol%)		0.05						0.10						0.15					
Case		1–1	1–2	1–3	2–1	2–2	2–3	1–1	1–2	1–3	2–1	2–2	2–3	1–1	1–2	1–3	2–1	2–2	2–3
Pressure ratio	PR_{comp1}	3.0	3.0	3.0	2.20	1.87	2.30	3.0	3.0	3.0	2.27	2.59	2.05	3.0	3.0	3.0	2.40	2.79	3.0
	PR_{comp2}	1.79	1.56	1.80	1.01	1.40	1.01	1.84	1.65	1.22	1.01	1.01	1.01	1.80	1.86	1.01	1.07	1.01	1.01
	PR_{comp3}	3.0	3.0	2.17	2.90	2.79	2.2	2.98	3.0	2.81	3.0	2.01	2.77	2.97	3.0	2.99	2.75	2.62	2.02
Pressure (bar)	P_{R5}	–	–	12.01	–	–	10.47	–	–	12.0	–	–	10.56	–	–	12.37	–	–	11.85
	P_{R6}:	1.77	1.71	2.47	2.99	3.45	3.17	1.87	1.50	1.66	2.76	3.21	2.43	1.74	1.49	1.41	2.79	3.35	3.62
Temperature (°C)	T_{R1}	32	32	–	32	32	–	32	31.9	–	32	−7.2	–	32	32	–	32	−6.16	–
	T_{R5}	–	–	−18.8	–	–	−0.79	–	–	−30.0	–	–	−3.6	–	–	−36.05	–	–	−2.9
Purge ratio	Purge1	0.0	0.02	0.02	0.0	0.0	0.0	0.03	0.03	0.06	0.0	0.0	0.0	0.07	0.05	0.19	0.0	0.0	0.0
	Purge2	–	0.0	0.0	–	0.0	0.0	–	0.0	0.0	–	0.0	0.0	–	0.0	0.0	0.0	0.0	0.0
Refrigerant composition (mol%)	C_{C2}	–	–	6.28	–	–	0.0	–	–	8.35	–	–	0.0	–	–	8.24	–	–	0.0
	C_{C3}	–	–	71.96	–	–	79.6	–	–	63.03	–	–	83.14	–	–	67.17	–	–	94.9
	C_{iC4}	–	–	10.84	–	–	17.63	–	–	15.56	–	–	2.72	–	–	15.90	–	–	2.15
	C_{nC4}	–	–	10.92	–	–	2.77	–	–	13.06	–	–	14.14	–	–	8.69	–	–	2.96
CO_2 emission (ton)		27.3	26.3	24.8	12.0	11.9	11.5	32.7	29.4	25.8	12.9	12.6	12.4	37.8	31.7	27.3	13.8	13.7	13.7

Table A. 3
Main stream results: N_2 0.05 mol% in LCO_2 at laden voyage

Case	Case 1-1			Case 1-2			Case 1-3			Case 2-1			Case 2-2			Case 2-3		
Stream	Press	Temp	Vapor	Press	Temp	Vapor	Press	Temp	Vapor	Press	Temp	Vapor	Press	Temp	Vapor	Press	Temp	Vapor
	(bar)	(°C)	Fraction	(bar)	(°C)	Fraction	(bar)	(°C)	Fraction	(bar)	(°C)	Fraction	(bar)	(°C)	Fraction	(bar)	(°C)	Fraction
S1	6.5	−40	1.0	6.5	−40	1.0	6.5	−40	1.0	15	−20	1.0	15	−20	1.0	15	−20	1.0
S5	47.13	121.9	1.0	58.42	118.9	1.0	30.61	66.52	1.0	48.78	49.76	1.0	44.99	67.73	1.0	30.02	24.16	1.0
S7	47.13	−13.7	0.02	58.42	−21.79	0.24	30.61	−34.24	0.17	48.78	−21.99	0.15	44.99	−21.96	0.07	30.02	−36.44	0.18
S7-1	–	–	–	19.5	−41.3	0.39	19.5	−41.3	0.23	–	–	–	44.99	−21.96	0.07	30.02	−36.44	0.18
S8	6.5	−55	0.31	6.5	−55	0.09	6.5	−55	0.09	15	−43.11	0.31	15	−38.15	0.15	15	−42.28	0.06
R1	2.17	32	1.0	1.51	32	1.0	2.69	31.44	1.0	1.49	31.81	1.0	1.5	32	1.0	1.48	24.88	1.0
R5	13.41	35	0.0	13.41	35	0.0	18.18	−30.28	0.0	13.41	35	0.0	13.41	35	0.0	11.92	−36.42	0.0
R6	2.17	−16.7	0.19	1.51	−24.79	0.21	2.69	−37.24	0.05	1.49	−24.99	0.21	1.5	−24.96	0.21	1.48	−39.44	0.02

Table A. 4
Main stream results: N_2 0.1 mol% in LCO_2 at laden voyage

Case	Case 1-1			Case 1-2			Case 1-3			Case 2-1			Case 2-2			Case 2-3		
Stream	Press	Temp	Vapor	Press	Temp	Vapor	Press	Temp	Vapor	Press	Temp	Vapor	Press	Temp	Vapor	Press	Temp	Vapor
	(bar)	(°C)	Fraction	(bar)	(°C)	Fraction	(bar)	(°C)	Fraction	(bar)	(°C)	Fraction	(bar)	(°C)	Fraction	(bar)	(°C)	Fraction
S1	6.5	−40	1.0	6.5	−40	1.0	6.5	−40	1.0	15	−20	1.0	15	−20	1.0	15	−20	1.0
S5	37.73	99.29	1.0	57.44	119.2	1.0	29.43	65.44	1.0	64.31	104.1	1.0	68.8	66.14	1.0	38.62	31	1.0
S7	37.73	−9.42	0.34	57.44	−20.6	0.31	29.43	−34.32	0.22	64.31	−22.01	0.19	68.8	−22.02	0.10	38.62	−38.6	0.09
S7-1	–	–	–	19.2	−41.6	0.44	19.5	−41.2	0.27	–	–	–	45.0	−26.8	0.19	38.62	−38.6	0.09
S8	6.5	−55	0.54	6.5	−55	0.09	6.5	−55	0.09	15	−46.69	0.38	15	−40.37	0.14	15	−45.71	0.09
R1	2.6	32	1.0	1.59	32	1.0	2.70	31.17	1.0	1.49	32	1.0	1.49	32	1.0	2.24	29.23	1.0
R5	13.41	35	0.0	13.41	35	0.0	18.76	−28.19	0.0	13.41	35	0.0	13.41	35	0.0	16.73	−38.58	0.0
R6	2.6	−12.42	0.17	1.59	−23.6	0.21	2.70	−37.32	0.06	1.49	−25.01	0.21	1.49	−25.02	0.21	2.24	−41.60	0.02

Table A. 5
Main stream results: N_2 0.15 mol% in LCO_2 at laden voyage

Case	Case 1-1			Case 1-2			Case 1-3			Case 2-1			Case 2-2			Case 2-3		
Stream	Press	Temp	Vapor	Press	Temp	Vapor	Press	Temp	Vapor	Press	Temp	Vapor	Press	Temp	Vapor	Press	Temp	Vapor
	(bar)	(°C)	Fraction	(bar)	(°C)	Fraction	(bar)	(°C)	Fraction	(bar)	(°C)	Fraction	(bar)	(°C)	Fraction	(bar)	(°C)	Fraction
S1	6.5	−40	1.0	6.5	−40	1.0	6.5	−40	1.0	15	−20	1.0	15	−20	1.0	15	−20	1.0
												0						
S5	39.77	105.3	1.0	58.49	133.1	1.0	30.58	68.3	1.0	69.73	78	1.0	69.97	64.52	1.0	44.32	28.53	1.0
S7	39.77	−5.47	0.66	58.49	−21.86	0.24	30.58	−32.81	0.31	69.73	−22.03	0.24	69.97	−22.03	0.18	44.32	−40.79	0.14
S7-1	–	–	–	19.5	−41.2	0.38	19.5	−41.2	0.35	–	–	–	45.0	−27.72	0.26	44.3	−40.79	0.14
S8	6.5	−55	0.77	0.6.5	−55	0.09	6.5	−55	0.09	15	−48.37	0.43	15	−40.83	0.14	15	−48.21	0.1
R1	3.05	32	1.0	1.50	31.0	1.0	2.37	31.2	1.0	1.49	32	1.0	1.49	32	1.0	1.97	27.14	1.0
R5	13.41	35	0.0	13.41	35	0.0	17.07	−21.28	0.0	13.41	35	0.0	13.41	35	0.0	16.28	−39.92	0.0
R6	3.05	−8.47	0.16	1.50	−24.86	0.21	2.37	−35.81	0.1	1.49	−25.03	0.21	1.49	−25.03	0.21	1.97	−43.79	0.03

Table A. 6
Calculated results for cost estimation of CO_2 BOG re-liquefaction processes

Case		Case 1-1			Case 1-2			Case 1-3			Case 2-1			Case 2-2			Case 2-3		
Storage Pressure	bar	6.5			6.5			6.5			15			15			15		
N_2 in LCO_2	mol%	0.05	0.1	0.15	0.05	0.1	0.15	0.05	0.1	0.15	0.05	0.1	0.15	0.05	0.1	0.15	0.05	0.1	0.15
1)CAPEX																			
Cargo Tank	M$	16.68	16.68	16.68	16.68	16.68	16.68	16.68	16.68	16.68	22.50	22.50	22.50	22.50	22.50	22.50	22.50	22.50	22.50
Except cargo tank	M$	5.37	5.20	5.04	5.93	5.83	5.82	5.42	5.42	5.38	5.44	5.45	5.63	5.35	5.64	5.87	4.94	5.0	5.04
2)Annualized CAPEX																			
Cargo Tank	M$/yr	1.96	1.96	1.96	1.96	1.96	1.96	1.96	1.96	1.96	2.64	2.64	2.64	2.64	2.64	2.64	2.64	2.64	2.64
Except cargo tank	M$/yr	0.63	0.61	0.59	0.70	0.68	0.68	0.64	0.64	0.63	0.64	0.64	0.66	0.63	0.66	0.69	0.58	0.59	0.59
Annualized cost	M$/yr	2.59	2.57	2.55	2.66	2.64	2.64	2.60	2.60	2.59	3.28	3.28	3.30	3.27	3.31	3.33	3.22	3.23	3.24
3)OPEX	M$/yr	0.11	0.09	0.08	0.16	0.16	0.14	0.13	0.13	0.12	0.10	0.12	0.12	0.09	0.10	0.10	0.08	0.09	0.10
4)TAC	M$/yr	2.70	2.66	2.63	2.82	2.80	2.79	2.72	2.72	2.71	3.39	3.40	3.43	3.36	3.41	3.44	3.31	3.32	3.34
5) Tax_{CO2}	M$/yr	0.36	0.68	1.01	0.16	0.23	0.30	0.14	0.21	0.29	0.07	0.10	0.12	0.07	0.08	0.10	0.06	0.07	0.08
6)Transport quantity	10^3ton/yr	320.4	319.2	317.9	321.9	322.4	322.9	321.9	322.4	322.9	297.4	297.7	298.0	297.4	297.7	298.1	297.4	297.8	298.2
7)CPT	$/$ton_{CO2}$	9.57	10.45	11.46	9.25	9.40	9.56	8.90	9.10	9.29	11.63	11.77	11.91	11.53	11.73	11.87	11.31	11.38	11.45

APPENDIX B. Economic comparison of the re-liquefaction process and the venting case

Fig.B. 1 compared the CPT between the worst re-liquefaction cases (Case 1-1, 2-1) and case where all BOG was vented while paying the carbon tax. At 6.5 bar storage pressure, venting case became more expensive than re-liquefaction case if the carbon tax exceeded 150 USD/ton$_{CO2}$. At 15 bar storage pressure, venting case became more expensive when the carbon tax exceeded 100 USD/ton$_{CO2}$. Considering the carbon tax of 150 USD/ton$_{CO2}$ in this study, all re-liquefaction processes are shown to be more economical compared to the case of venting CO_2 BOG.

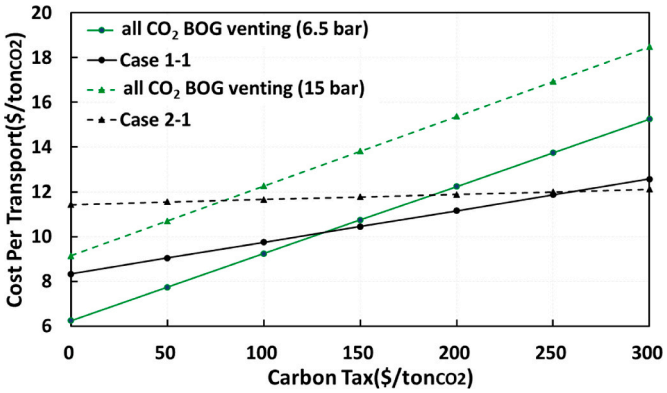


Fig. B. 1. Economic comparison of re-liquefaction and venting cases at 0.1 mol% N_2 with carbon tax from 0 to 300 USD.

Data availability

Data will be made available on request.

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